

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Summative Assessment 1

Summative Assessment (SA-1)

Class: Class 10

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Networking Concepts, Internet and Web Technologies, HTML
Fundamentals, CSS Styling

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

- a. Which layer of the OSI model is responsible for end-to-end communication?
 - b. Physical layer
 - c. Data link layer
 - d. Network layer
 - e. Transport layer
- b. Which of the following is a Class C IP address?
 - c. 10.0.0.1
 - d. 172.16.0.1
 - e. 192.168.1.1
 - f. 224.0.0.1
- c. The `<!DOCTYPE html>` declaration in HTML5:
 - d. Creates a paragraph
 - e. Tells the browser to render in quirks mode
 - f. Tells the browser this is an HTML5 document
 - g. Is optional and can be removed
- d. Which CSS property creates space between the element border and its content?
 - e. margin
 - f. padding
 - g. border-spacing
 - h. gap
- e. Which HTML5 semantic tag represents a self-contained composition?
 - f. `<div>`
 - g. `<section>`
 - h. `<article>`
 - i. `<aside>`
- f. What is the purpose of `meta charset="UTF-8"` in HTML?

- g. Sets the page title
- h. Defines character encoding
- i. Adds viewport settings
- j. Links external stylesheet
- g. Which HTTP status code indicates “Page Not Found”?
- h. 200
- i. 301
- j. 404
- k. 500
- h. In CSS, which selector has the **highest** specificity?
- i. Element selector p
- j. Class selector .box
- k. ID selector #main
- l. Universal selector *
- i. Which tag is used to embed a video in HTML5?
- j. <video>
- k. <embed>
- l. <media>
- m. <movie>
- j. The position: sticky CSS property:
- k. Positions element relative to the viewport
 - l. Positions element relative to its normal position until scroll threshold
- m. Positions element relative to parent
- n. Removes element from document flow

Part B: Short Answer (8 x 2 = 16 marks)

- a. Explain the difference between **IPv4** and **IPv6**. Why was IPv6 introduced? Give the format of each.
- b. Write the HTML code for a form that collects: Full Name, Email, Phone (with pattern validation), Date of Birth, Gender (radio buttons), and a Submit button. Include proper labels and validation attributes.
- c. Differentiate between **block-level** and **inline** elements with three examples each. Which elements are inline-block?
- d. What is the 盒模型 (box model) in CSS? Explain content, padding, border, and margin with a labelled diagram.
- e. Explain the TCP/IP model with its four layers. Name two protocols at each layer.
- f. Write CSS rules to:

- Create a responsive navigation bar with flexbox
 - Style links with hover underline effect
 - Set a media query for screens below 768px width
 - Create a 3-column grid layout
- g. What is **responsive web design**? Explain three techniques used to achieve it (media queries, flexbox, relative units).
- h. Differentiate between class and id selectors in CSS. Explain specificity rules with examples.

Part C: Long Answer (4 x 3 = 12 marks)

- a. Create a complete HTML page for a **Student Management System** that includes:
- A header with school name and logo placeholder
 - A form to add new students (name, roll, class, section, marks)
 - A styled table displaying 5 students with sorting capability
 - CSS for professional formatting with borders, colours, spacing
 - Footer with contact information

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- b. Write a detailed comparison of **Bus**, **Star**, and **Mesh** topologies. For each, explain:
- Structure with a diagram
 - How data flows
 - Two advantages and two disadvantages
 - Best use case scenario

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- c. Create a complete CSS-styled webpage for a **Restaurant Menu** that includes:
- Hero section with restaurant name and tagline
 - Menu section with food items in categories (Starters, Main Course, Desserts)
 - Use CSS Grid for layout
 - Responsive design that works on mobile
 - Proper typography and colour scheme
- d. Explain **DNS resolution process** step by step. Include:
- What is DNS and why it is needed
 - Types of DNS servers (recursive, root, TLD, authoritative)
 - The complete resolution process with a diagram
 - What happens when you type `www.example.com` in the browser

Part D: Application Based (2 x 1 = 2 marks)

- a. A student creates a webpage where the navigation bar disappears on mobile. The CSS code is:

```
nav {
  display: flex;
  justify-content: space-around;
}
@media (max-width: 768px) {
  nav {
    display: none;
  }
}
```

Identify the problem and write the corrected CSS that shows a hamburger menu on mobile. Explain your approach.

b. A company has three offices in Bangalore, Mysore, and Hubli. They want to:

- Connect all offices with high-speed data transfer
- Share a common database server
- Allow video conferencing between offices

Recommend the appropriate network type (LAN/MAN/WAN) and explain why. Suggest two protocols that would be used.